



Multivessel coronary artery bypass grafting via small thoracotomy versus sternotomy (MIST): an investigator-initiated, international, open-label, randomised controlled trial

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Summary

Background Coronary artery bypass grafting (CABG) is a common but invasive operation traditionally performed through a median sternotomy. Minimally invasive cardiac surgery (MICS) CABG via small thoracotomy might improve postoperative recovery, but randomised evidence has been scarce. We aimed to compare patient-reported recovery after MICS CABG versus sternotomy CABG in patients with multivessel coronary artery disease and to describe clinical and safety outcomes.

Methods MIST was an investigator-initiated, international, open-label, randomised controlled trial done at seven centres (four academic hospitals and three community hospitals) in Canada, India, China, Germany, the USA, and Japan. Patients referred to participating surgeons for CABG were eligible if they were aged 18 years or older; had angiographically confirmed multivessel coronary artery disease, defined as lesions of at least 70% stenosis in at least two major epicardial vessels and at least two separate coronary artery territories (left anterior descending artery, left circumflex artery, or right coronary artery) or left main coronary stenosis of 50% or more; and were suitable for coronary surgery both with sternotomy CABG and MICS CABG. Patients who were haemodynamically compromised, had contraindications to either approach, had had previous cardiac surgery, or required concomitant procedures were excluded. Eligible patients were randomly assigned (1:1) to MICS CABG or sternotomy CABG by a central, web-based system, stratified by centre, with block sizes of four and six. The primary endpoint was patient-reported physical recovery at 1 month, assessed by the 36-item Short Form Health Survey Physical Component Summary (SF-36 PCS) score. The primary analysis was by intention to treat; safety analyses were done according to treatment received. Missing 1-month questionnaire data were handled by multiple imputation. The trial was registered with ClinicalTrials.gov (NCT03447938), and is closed to recruitment.

Findings Between Aug 24, 2018, and Nov 26, 2024, 176 patients were enrolled, 170 of whom were randomly assigned to MICS CABG (n=86) or sternotomy CABG (n=84). The median age of patients was 67·0 years (IQR 61·0–72·0), 154 (91%) patients were male, and 16 (9%) were female. At 1 month after surgery, SF-36 PCS scores were significantly higher in the MICS CABG group than in the sternotomy CABG group (mean 45·1 [SD 8·0] vs 42·2 [9·1]; mean difference 2·9 [95% CI 0·3–5·5]; p=0·031). Clinical and safety follow-up at 1 month was complete in all patients; 12-month clinical and safety follow-up was complete in all except three patients in the sternotomy CABG group. Up to 12 months after surgery, there were no deaths or strokes in either group; one major adverse cardiac or cerebrovascular event occurred in the MICS CABG group before 1 month and none in the sternotomy CABG group.

Interpretation For selected patients with multivessel coronary artery disease, MICS CABG performed by experienced teams improved patient-reported physical recovery at 1 month compared with sternotomy CABG, with no apparent safety penalty through to 12 months. These findings support consideration of MICS CABG in appropriately selected patients treated by experienced teams, and further studies of implementation, recovery pathways, and long-term outcomes.

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Introduction

Coronary artery bypass grafting (CABG) is one of the most commonly performed major surgeries worldwide.^{1,2} Compared with percutaneous coronary intervention

(PCI), CABG can improve long-term outcomes, including reducing the risk of myocardial infarction and the need for repeat revascularisation, in selected patients with multivessel coronary artery disease with or without left

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Research in context

Evidence before this study

The invasiveness of coronary artery bypass grafting (CABG), a common major operation worldwide, has remained mostly unchanged since its introduction nearly 60 years ago. The median sternotomy incision is still considered the standard approach, but is associated with considerable morbidity, including infection, post-sternotomy pain, sternal malunion, and a slow return to full physical activity. Minimally invasive cardiac surgery (MICS) CABG was developed to allow surgical revascularisation of multiple myocardial territories through a small thoracotomy incision, often without cardiopulmonary bypass. We searched PubMed, Embase, and Cochrane Library with no language restrictions for articles published from database inception to June 1, 2026, with combinations of the terms “minimally invasive cardiac surgery”, “minimally invasive coronary* OR CABG”, “MICS CABG* OR CABG”, “robotic coronary* OR CABG”, “hybrid coronary* OR CABG”. The search was restricted to human studies and included observational studies, randomised controlled trials, and meta-analyses. Current literature has reported favourable short-term and long-term outcomes after MICS CABG, supporting its feasibility and suggesting low adverse-event rates in experienced centres. Several comparative observational studies have reported fewer transfusions and surgical site infections in patients who underwent MICS CABG compared with those who have sternotomy CABG. However, randomised evidence comparing contemporary MICS CABG with sternotomy CABG, including patient-reported recovery and clinical and safety outcomes, has been scarce.

Added value of this study

Although many surgical specialties have moved towards less invasive approaches, CABG largely has not. To our knowledge, the MIST trial is the first randomised controlled trial comparing MICS CABG with sternotomy CABG for multivessel coronary artery disease. Until now, the case for a less invasive, non-sternotomy approach has rested largely on single-centre series and observational registries, which are susceptible to selection and operator-experience biases. By generating randomised, multicentre evidence in a field where high-quality surgical trials remain uncommon, the trial helps clarify whether avoiding a

sternotomy can improve patient-reported physical recovery at 1 month while showing no apparent reduction in procedural quality or safety through to 12 months, concerns that have tempered wider adoption of MICS CABG. It also provides randomised evidence on patient-reported recovery, an outcome of direct relevance to patients that has been less commonly used in the evaluation of surgical revascularisation. Performed by experienced teams in appropriately selected patients, the MIST trial offers recovery, procedural, and safety data that can inform the design of future trials, surgical training, and implementation research around MICS CABG.

Implications of all the available evidence

For selected patients with multivessel coronary artery disease treated by experienced teams, MICS CABG improved patient-reported physical recovery at 1 month and was associated with higher physical domain scores, fewer transfusions, and shorter total ventilation time compared with sternotomy CABG, without an apparent penalty in procedural quality or 12-month safety outcomes. Clinical and safety events were infrequent, and the trial was not powered to compare clinical outcomes. Long-term follow-up of the MIST trial is ongoing and will ascertain whether early patient-centred benefits translate into durable long-term outcomes. These findings provide randomised evidence supporting the potential recovery advantages of MICS CABG in appropriately selected patients treated by experienced teams, and support further research, training, and implementation work. The relevance of these findings extends beyond the scientific community to patients and the public. The MIST trial used validated tools to evaluate patient-relevant outcomes, namely self-reported functional recovery and return to daily activities after CABG. It provides randomised evidence of improved patient-reported recovery with no apparent safety penalty through 12 months for a less invasive surgical revascularisation approach that avoids a sternotomy incision for patients with multivessel coronary artery disease. For patients, this suggests that, in appropriate circumstances and with experienced operators, multivessel bypass surgery may be achievable without a median sternotomy—an option whose role further clinical research will continue to define.

main involvement.^{2–6} The median sternotomy incision has been the standard approach for multivessel CABG for almost 60 years, despite drawbacks such as delayed return to full physical activity, risk of mediastinal infection, sternal malunion, and chronic pain.^{7–10} Conventional CABG can impose a substantial physical, functional, and psychosocial burden on patients. Previous trials comparing CABG with PCI suggest that recovery after CABG is slower than after PCI and that some recovery differences can persist up to 12 months.^{8,11}

Minimally invasive cardiac surgery (MICS) CABG was developed to allow multivessel surgical coronary

revascularisation through a small anterolateral thoracotomy incision, typically measuring 5 cm or less, with the aim of avoiding some of the drawbacks of sternotomy CABG and facilitating recovery.^{12,13} Observational studies have reported low operative mortality with MICS CABG (0–2%), favourable long-term functional outcomes, and 12-year survival of 82%.^{12–17} MICS CABG is increasingly being used in the USA;¹⁸ however, evidence supporting its benefit has been scarce, with most reports of technical success to date being limited to non-randomised, single-centre studies. We therefore aimed to compare patient-reported physical

recovery after MICS CABG versus sternotomy CABG in patients with multivessel coronary artery disease and to assess health-related quality of life, procedural outcomes, and clinical and safety outcomes.¹⁹

Methods

Study design and participants

MIST was an investigator-initiated, international, open-label, randomised controlled trial comparing MICS CABG through a small thoracotomy with CABG through sternotomy in patients undergoing isolated CABG for multivessel coronary artery disease. The trial design and rationale have been previously described.¹⁹

The trial was conducted at seven centres (four academic and three community hospitals) across six countries: Canada (two centres), India, China, Germany, the USA, and Japan. A list of participating sites and investigators is in the appendix (p 2). Patients referred to participating surgeons for CABG were eligible if they were aged 18 years or older; had angiographically confirmed multivessel coronary artery disease, defined as stenosis of at least 70% in at least two major epicardial vessels and at least two separate coronary artery territories (left anterior descending artery, left circumflex artery, or right coronary artery), or left main coronary stenosis of 50% or more; were suitable for coronary surgery by either CABG via sternotomy or MICS CABG; and were willing to attend all follow-up study visits.¹⁹ Patients were excluded if they were younger than 18 years; had a concomitant cardiac procedure with CABG; had previous cardiac surgery, mediastinal irradiation, or clinically significant trauma to the chest; had contraindications for MICS CABG (severe pectus excavatum or pulmonary disease, haemodynamically significant left subclavian stenosis, severe obesity, severe left ventricular dysfunction, no adequate posterior descending artery or marginal branch target, and absence of femoral pulse bilaterally); had contraindications to CABG via sternotomy (contraindication to both off-pump and on-pump CABG); had a history of breast augmentation; had concomitant life-threatening disease likely to limit life expectancy to less than 2 years; had emergency CABG with haemodynamic compromise; or were unable to provide informed consent. Full enrolment criteria are provided in the protocol.¹⁹

Patient sex was self-reported as male or female. Race and ethnicity data were not collected given the multinational nature of the trial, as definitions and classifications of race and ethnicity varied by region. Written informed consent was obtained from all patients.

The protocol (appendix pp 25–44) was approved by the institutional review board at University of Ottawa Heart Institute (20180008–01H) and subsequently at each study site (appendix p 3). The trial was conducted in accordance with the principles of the Declaration of Helsinki. Complete details about the trial design, statistical analyses, trial oversight, and procedures are provided in the study protocol.¹⁹ This trial was prospectively registered

on ClinicalTrials.gov (NCT03447938) and is closed to new patients; follow-up is ongoing.

A multidisciplinary steering committee led the oversight of the trial. An independent data and safety monitoring committee reviewed adverse events, safety considerations, and data integrity. Patients were engaged through focus groups, which informed the trial's design and reporting. One coauthor is a patient who was closely involved from the inception of the trial design. The trial's design was also informed by our group's previous research on patient preferences following cardiac surgery.²⁰

Randomisation and masking

Patients were identified by surgeons at participating centres and screened and enrolled by institutional research coordinators. Patients were randomly assigned (1:1) to MICS CABG (small thoracotomy) or sternotomy CABG by means of a central, web-based randomisation system (Dacima Clinical Suite version 4.3.5; EvidentIQ, Montreal, QC, Canada), with concealed allocation and no access to the next assignment in the sequence. Block randomisation was used, with permuted block sizes of four and six, stratified by centre. Investigators and patients were not masked to treatment allocation. The research coordinators who performed the follow-up visits were masked to treatment allocation, but success of this masking was not formally evaluated.

See Online for appendix

Procedures

Participating surgeons were required to be experienced with both study procedures, defined as having more than 3 years of independent practice, and having performed at least 100 multivessel MICS CABG and at least 500 multivessel sternotomy CABG operations.

MICS CABG was performed through a small anterolateral thoracotomy (typically ≤ 5 cm), with myocardial exposure achieved using a positioner and an epicardial stabiliser, with or without an additional 8 mm subxiphoid incision. The detailed surgical technique has been previously described.^{12,13,19,21} Sternotomy CABG was performed via standard median sternotomy. For all patients, graft number, type (arterial or venous), and configuration were determined by the surgeon preoperatively. Use of cardiopulmonary bypass, graft harvesting technique, and robotic assistance were left to the surgeon's discretion. No specific surgical equipment was mandated, and patients were otherwise monitored and treated according to the local standard of care. The participating surgeon at each institution performed both MICS CABG and sternotomy CABG for the trial patients.

Study investigators followed up all patients to collect clinical outcomes during the index hospital admission (from surgery until discharge). Follow-up data were obtained by research coordinators at in-person visits 1 month after surgery, with additional clinical visits at 3, 6, and 12 months after surgery (appendix pp 12–14).

Outcomes

All outcomes were centrally assessed. The primary endpoint was patient-reported physical recovery at 1 month after surgery, assessed with the 36-item Short Form Health Survey (SF-36) Physical Component Summary (PCS) score. The SF-36 has been used to quantify functional status in postoperative and other clinical settings.^{22–25}

The secondary efficacy endpoints included individual SF-36 domains: physical functioning; role-physical (role limitations due to physical health problems); bodily pain; general health; vitality; social functioning; role-emotional (role limitations due to emotional problems); and mental health. The SF-36 PCS and Mental Component Summary (MCS) were derived from these domains.

A secondary safety endpoint was the first occurrence of a major adverse cardiac or cerebrovascular event (MACCE), a composite of all-cause death, myocardial infarction, stroke, or target vessel revascularisation. Each component of the composite endpoint was also evaluated individually. Additional secondary outcomes included number of grafts per patient, proportion of arterial grafts, postoperative transfusion requirement, duration of mechanical ventilation, length of hospital stay, postoperative pain by the visual analogue scale (VAS; resting and post-physiotherapy), freedom from angina (assessed by the Seattle Angina Questionnaire [SAQ]), and quality of life assessed by the EQ-5D VAS component of the EQ-5D 5-level questionnaire and SF-36 MCS scores. Other prespecified secondary endpoints were index of completeness of revascularisation, types of

transfusion, incidence of re-exploration for bleeding, atrial fibrillation, wound infection, and length of intensive care unit stay (appendix pp 11–18). Efficacy and safety endpoints were assessed at 1, 3, 6, and 12 months after surgery. Outcomes at 3, 6, and 12 months will be reported elsewhere.

Statistical analysis

A sample size of 176 patients provided 80% power to detect a mean between-group difference of 5 points in the SF-36 PCS score at 1 month after surgery, assuming an SD of 10, a two-sided significance level of 5%, and allowing for estimated withdrawal and crossover rates of 10% each. No interim analyses were planned.

The primary analysis was done in the intention-to-treat population; safety analyses were done according to treatment received. Per-protocol analyses were used as sensitivity analyses. Descriptive statistics for baseline characteristics and endpoints included proportions, means (SDs), or medians (IQRs). For the short-term, 1-month analysis, the primary endpoint of SF-36 PCS score was compared using the *t* test, and the mean difference and 95% CI were calculated. As a sensitivity analysis, a multiple regression model was used to adjust for possible differences in the key baseline covariates of age, sex, use of cardiopulmonary bypass, number of grafts, and BMI between the study groups. In an additional sensitivity analysis, between-group differences were assessed with the Wilcoxon rank-sum test. For the secondary endpoints, mean differences with 95% CIs were calculated for the eight SF-36 domains, SF-36 MCS score, SAQ summary score, and EQ-5D VAS. Multiple imputation was used to handle missing data for 1-month outcomes. MACCEs, target vessel revascularisation, and other secondary endpoints were evaluated with risk ratios (RRs) and 95% CIs calculated for discrete endpoints, and mean differences and 95% CI for continuous endpoints, when possible.

For non-normally distributed continuous variables, the groups were compared using the Wilcoxon rank-sum test and quantile regression was used to estimate the median difference between groups with the corresponding 95% CI. Similarly, trial interventions and operative characteristics were compared between groups and RR, mean difference, and median differences with 95% CIs were calculated. Exploratory subgroup analyses were performed according to sex, the presence or absence of cardiovascular risk factors (including hypertension, dyslipidaemia, and diabetes), and previous myocardial infarction or previous PCI. For the long-term analysis of SF-36 component and domains scores, SAQ summary score, and EQ-5D, a 2×4 mixed-model repeated-measures analysis was conducted with the between-factor group and the within-factor time (1, 3, 6, and 12 months) and the tests for group-by-time, group, and time conducted. Missing data were handled within the mixed-model repeated measures. A two-sided *p* value of less than 0.05 was considered statistically

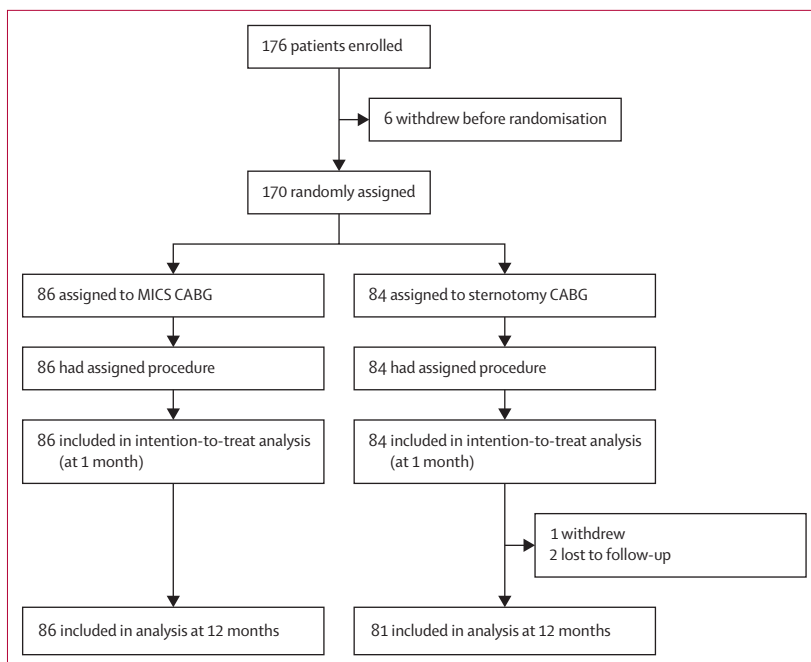


Figure 1: Trial profile

CABG=coronary artery bypass grafting. MICS CABG=minimally invasive cardiac surgery CABG.

significant. All analyses were performed with SAS software, version 9.4.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between Aug 24, 2018, and Nov 26, 2024, 176 patients were enrolled, 82 (47%) of whom were enrolled during the last 2 years of the trial, largely due to COVID-19 disruptions. Each centre's enrolment period and the number of operations performed by the participating surgeons are shown in the appendix (p 2). Six patients voluntarily withdrew their consent before randomisation; 86 patients were randomly assigned to MICS CABG and 84 patients to sternotomy CABG (figure 1). 16 (9%) of 170 patients were female and 154 (91%) were male. The median age was 66·5 years (IQR 60·0–70·0) in the MICS CABG group and 68·5 years (62·0–74·0) in the sternotomy CABG group (table 1). A left ventricular ejection fraction of less than 50% was present in 28 (16%) of 170 patients, and 40 (24%) presented with unstable angina. Three-vessel coronary artery disease or left main stenosis with additional coronary artery involvement was present in 140 (82%) of 170 patients.

There were no crossovers between the two groups; no patients in the MICS CABG group required conversion to sternotomy (figure 1). There was no difference between groups in the total number of grafts or in the number of distal arterial anastomoses (table 2). MICS CABG was associated with a longer procedural duration than sternotomy CABG and with shorter mechanical ventilation time (table 2). Transfusion of any type was needed in fewer patients in the MICS CABG group than in the sternotomy CABG group (11 [13%] of 85 patients in the MICS CABG group vs 27 [32%] of 84 in the sternotomy CABG group; RR 0·40 [95% CI 0·21–0·76]).

At 1 month after surgery, the mean SF-36 PCS score was higher in the MICS CABG group than in the sternotomy CABG group (mean 45·1 [SD 8·0] vs 42·2 [9·1]; mean difference 2·9 [95% CI 0·3–5·5]; $p=0·031$; table 3). Findings were similar after adjustment for centre and for key covariates, including age, sex, use of cardiopulmonary bypass, number of grafts, and BMI (table 3). Exploratory subgroup analyses did not show significant treatment interactions between the relative effects of the two procedures across subgroups (appendix p 21).

In addition to the PCS score, the SF-36 role-physical and physical functioning domain scores were higher in the MICS CABG group than in the sternotomy CABG group 1 month after surgery (figure 2; table 3). No significant between-group differences were identified for the bodily pain or general health domain scores (table 3).

At 1 month after surgery, no significant between-group differences were identified in EQ-5D VAS, SAQ summary, or SF-36 MCS scores (table 3). No clear

between-group differences were identified in the SF-36 mental domain scores (figure 2; table 3). The proportion of arterial grafts per patient, index of completeness of revascularisation, postoperative pain scores, intensive

	MICS CABG group (n=86)	Sternotomy CABG group (n=84)
Age, years	66·5 (60·0–70·0)	68·5 (62·0–74·0)
Sex*		
Male	79 (92%)	75 (89%)
Female	7 (8%)	9 (11%)
BMI, kg/m ²	26·4 (23·7–28·6)	26·8 (23·5–28·9)
Diabetes	43 (50%)	42 (50%)
Insulin-dependent	6 (14%)	7 (17%)
Non-insulin-dependent†	37 (86%)	35 (83%)
Hypertension	67 (78%)	62 (74%)
Dyslipidaemia	44 (51%)	48 (57%)
Smoking status (ever)	31 (36%)	31 (37%)
Current tobacco user	15 (48%)	7 (23%)
Former tobacco user	16 (52%)	24 (77%)
Never smoker	55 (64%)	53 (63%)
Previous myocardial infarction	29 (34%)	22 (26%)
Previous percutaneous coronary intervention	18 (21%)	14 (17%)
Congestive heart failure	7 (8%)	2 (2%)
History of transient ischaemic attack or cerebrovascular accident	6 (7%)	6 (7%)
COPD	4 (5%)	1 (1%)
Peripheral vascular disease	4 (5%)	5 (6%)
Renal failure requiring dialysis	1 (1%)	2 (2%)
Atrial fibrillation	3 (3%)	4 (5%)
LVEF <50%	13 (15%)	15 (18%)
Unstable angina	16 (19%)	24 (29%)
Urgent indication for surgery (same admission)	11 (13%)	10 (12%)
Antiplatelet agents before surgery	68 (79%)	65 (77%)
Number of diseased vessels	3·0 (3·0–3·0)	3·0 (3·0–3·0)
Diseased vessels		
Left main only	0	0
Double-vessel disease	15 (17%)	15 (18%)
Triple-vessel disease	43 (50%)	38 (45%)
Left main plus single-vessel disease	5 (6%)	5 (6%)
Left main plus double-vessel disease	11 (13%)	13 (15%)
Left main plus triple-vessel disease	12 (14%)	13 (15%)

Data are median (IQR) or n (%). CABG=coronary artery bypass grafting. COPD=chronic obstructive pulmonary disease. LVEF=left ventricular ejection fraction. MICS CABG=minimally invasive cardiac surgery CABG. *One transgender woman self-categorised as female. †Includes diet control, oral medication, or self-reported non-adherence to diet control (one patient in each group reported non-adherence to diet control).

Table 1: Baseline characteristics

	MICS CABG group (n=86)	Sternotomy CABG group (n=84)	MD, MeD, or RR (95% CI)
Time to procedure, * days	4.5 (1.0–31.0)	7.5 (2.0–29.0)	MeD -2.9 (-8.5 to 2.6)
Number of grafts	3.0 (2.0–3.0)	3.0 (2.0–3.0)	MeD 0.0 (-0.4 to 0.4)
Number of distal arterial anastomoses	2.0 (1.0–2.0)	2.0 (1.0–3.0)	MeD 0.0 (-0.4 to 0.4)
Conduits used			
LITA use	86 (100%)	84 (100%)	..
Robotic harvesting†	5 (6%)	0	..
Skeletonised	62 (72%)	54 (64%)	RR 1.12 (0.91 to 1.38)
RITA use	19 (22%)	15 (18%)	RR 1.23 (0.67 to 2.27)
Radial use	20 (23%)	23 (27%)	RR 0.85 (0.51 to 1.43)
Gastroepiploic artery use	1 (1%)	1 (1%)	RR 0.98 (0.06 to 15.36)
Multiple arterial distal anastomoses	40 (47%)	39 (46%)	RR 1.00 (0.73 to 1.38)
Arterial grafts only	22 (26%)	24 (29%)	RR 0.90 (0.55 to 1.47)
Grafts configuration			
In-situ LITA	86 (100%)	83 (99%)	..
Proximal aortic	62 (72%)	69 (82%)	RR 0.88 (0.74 to 1.04)
Y or T or I based	21 (24%)	12 (14%)	RR 1.71 (0.90 to 3.25)
Sequential LITA use	4 (5%)	3 (4%)	RR 1.30 (0.30 to 5.64)
Use of transit-time flowmetry	76 (88%)	72 (86%)	RR 1.03 (0.92 to 1.16)
Cardiopulmonary bypass			
Off-pump surgery	82 (95%)	84 (100%)	..
Time on cardiopulmonary bypass, min	103 (41–172); n=4	..	..
Left atrial appendage exclusion	0	1 (1%)	..
Procedure duration, min	219.5 (78.4)	181.4 (54.8)	MD 38.2 (17.6 to 58.7)
Total ventilation time, h	5.2 (5.3)	11.6 (38.5)	MD -6.4 (-14.7 to -1.9)

Data are median (IQR), n (%), or mean (SD), unless otherwise specified. CABG=coronary artery bypass grafting. LITA=left internal thoracic artery. MeD=median difference. MD=mean difference. MICS CABG=minimally invasive cardiac surgery CABG. RITA=right internal thoracic artery. RR=risk ratio. *Denotes the number of days from randomisation to surgery. †Robotic internal thoracic artery harvesting was used in five patients in the MICS CABG group at two of the centres.

Table 2: Trial interventions and operative characteristics

care unit stay, and hospital stay did not differ significantly between groups (table 3). Longitudinal quality-of-life analyses through to 12 months after surgery are reported in the appendix (pp 19–20).

Clinical and safety follow-up was complete for all patients at 1 month after surgery and for all except three patients at 12 months (one withdrawal and two lost to follow-up in the sternotomy CABG group). Serious adverse events from surgery to 1 month and from 1 month to 12 months after surgery are shown in table 4. One MACCE (postoperative catheterisation and rethoracotomy for a kinked graft) was reported before 1 month in the MICS CABG group and none in the sternotomy CABG group. During the 12 months after surgery, there were no deaths and no strokes in either group, and other serious adverse events were infrequent in both groups.

Discussion

In this multicentre, open-label, randomised controlled trial of patients with multivessel coronary artery disease,

MICS CABG performed by experienced surgeons improved patient-reported physical recovery at 1 month after surgery compared with sternotomy CABG, with no apparent reduction in procedural quality and no apparent penalty in safety through 12 months. MICS CABG was associated with shorter total ventilation time and fewer transfusions, findings consistent with potential perioperative advantages of a less invasive surgical revascularisation approach in appropriately selected patients. Specifically, patients assigned to MICS CABG had higher SF-36 PCS scores at 1 month than those assigned to sternotomy CABG, an observation consistent with earlier physical recovery. The SF-36 role-physical and physical functioning domain scores were also higher in the MICS CABG group. Clinical and safety events were infrequent through 12 months, with no deaths or strokes in either group, although the trial was not powered to compare clinical outcomes. The number of grafts, index of completeness of revascularisation, and arterial graft configuration did not differ between groups. Procedure duration was a mean of 38.2 min (95% CI 17.6–58.7) longer with MICS CABG than with sternotomy CABG.

To our knowledge, MIST is the first randomised controlled trial to show improved patient-reported early physical recovery with a minimally invasive approach to multivessel CABG compared with sternotomy. These results extend previous randomised evidence from mitral valve surgery to surgical coronary revascularisation. As in the previous UK Mini Mitral trial,²⁵ which compared mitral valve repair via a mini-thoracotomy versus sternotomy, the aim of the MIST trial was not to test superiority for hard clinical outcomes, but to assess whether a less invasive approach could improve recovery while maintaining procedural quality and monitoring safety. However, the UK Mini Mitral trial did not show a difference in the physical function T score at 12 weeks.²⁵ One key distinction between the UK Mini Mitral trial and the MIST trial is the timing of the primary endpoint. In the UK Mini Mitral trial, SF-36 physical function score and accelerometer data at 6 weeks postoperatively favoured the mini-thoracotomy group, but the 12-week primary endpoint showed no significant difference between the two groups.²⁶ Another difference between the two trials is that all but four procedures in the MIST trial were performed without cardiopulmonary bypass, whereas cardiopulmonary bypass is required for mitral surgery. In this regard, cardiopulmonary bypass and aortic cross-clamp times were longer and groin complications more frequent in the mini-thoracotomy group versus the sternotomy group of the UK Mini Mitral trial, which might have adversely affected early recovery and function.

The MIST trial suggests earlier functional recovery in the first month with a minimally invasive approach to multivessel CABG, possibly enabling faster return to daily activities for patients, in addition to being associated with fewer transfusions and shorter total ventilation time.

	MICS CABG group (n=86)*	Sternotomy CABG group (n=84)*	MD, MeD, or RR (95% CI)	p value
Primary outcome				
Primary analysis				
SF-36 PCS score	45.1 (8.0)	42.2 (9.1)	MD 2.9 (0.3 to 5.5)	0.031†
Sensitivity analysis				
SF-36 PCS score adjusting for centre	..	..	MD 2.9 (0.6 to 5.2)	0.015
SF-36 PCS score adjusting for covariates‡	..	..	MD 3.1 (0.5 to 5.7)	0.020
Secondary outcomes				
Role-physical	65.6 (28.5)	55.4 (31.3)	MD 10.2 (1.1 to 19.3)	0.028
Physical functioning	70.7 (19.7)	63.8 (19.1)	MD 6.9 (1.1 to 12.8)	0.021
Bodily pain	68.4 (19.8)	63.4 (21.6)	MD 5.0 (-1.2 to 11.3)	0.11
General health	72.5 (17.2)	72.6 (18.6)	MD -0.1 (-5.5 to 5.4)	0.98
Vitality	62.6 (17.3)	62.3 (19.0)	MD 0.4 (-5.1 to 5.9)	0.89
Social functioning	78.1 (21.1)	75.4 (21.0)	MD 2.6 (-3.8 to 9.0)	0.42
Role-emotional	84.2 (21.4)	79.9 (24.2)	MD 4.3 (-2.6 to 11.3)	0.22
Mental health	79.6 (13.1)	75.5 (13.7)	MD 4.1 (-0.004 to 8.1)	0.050
Postoperative pain VAS (resting)	3.2 (1.7)	3.5 (1.7)	MD -0.4 (-0.9 to 0.2)	0.19
Postoperative pain VAS (post-physiotherapy)	3.2 (1.9)	3.4 (2.0)	MD -0.2 (-0.8 to 0.4)	0.42
Death	0	0	..	..
Major adverse cardiovascular events	1 (1%)	0	..	..
Target vessel revascularisation	1 (1%)	0	..	..
Percentage of arterial grafts per patient	54.2 (27.6)	54.7 (29.9)	MD -0.5 (-9.2 to 8.2)	0.91
Index of completeness of revascularisation	0.8 (0.3)	0.8 (0.3)	MD -0.001 (-0.084 to 0.082)	0.98
Any red blood cell transfusion, intraoperative	4/84 (5%)	13/84 (16%)	RR 0.31 (0.10 to 0.91)	0.032
Any transfusion, intraoperative	6/84 (7%)	15/84 (18%)	RR 0.40 (0.16 to 0.98)	0.045
Any red blood cell transfusion, postoperative	6/85 (7%)	15/84 (18%)	RR 0.40 (0.16 to 0.97)	0.043
Any transfusion, postoperative	8/85 (9%)	17/84 (20%)	RR 0.47 (0.21 to 1.02)	0.056
Any red blood cell transfusion	8/85 (9%)	23/84 (27%)	RR 0.34 (0.16 to 0.72)	0.0050
Any transfusion	11/85 (13%)	27/84 (32%)	RR 0.40 (0.21 to 0.76)	0.0048
Incidence of re-exploration for bleeding	1/86 (1%)	3/84 (4%)	RR 0.32 (0.03 to 3.12)	0.33
Incidence of atrial fibrillation	5/86 (6%)	6/84 (7%)	RR 0.81 (0.26 to 2.57)	0.73
Incidence of wound infection	0/86	3/84 (4%)	..	..
Length of intensive care unit stay, days	1.0 (1.0-2.0)	1.0 (1.0-2.0)	MeD 0.0 (-0.4 to 0.4)	0.42§
Length of hospital stay, days	5 (4-8)	5 (4-7)	MeD -0.0 (-1.3 to 1.3)	0.087§
EQ-5D VAS score	78.2 (16.5)	77.9 (16.4)	MD 0.3 (-4.7 to 5.3)	0.90
SAQ score	80.4 (13.3)	80.7 (10.9)	MD -0.2 (-3.9 to 3.4)	0.89
MCS score	51.9 (7.9)	51.3 (8.5)	MD 0.7 (-1.8 to 3.2)	0.60

Data are mean (SD), n/N (%), or median (IQR), except where otherwise specified. CABG=coronary artery bypass grafting. MeD=median difference. MCS=Mental Component Summary. MD=mean difference. MICS CABG=minimally invasive cardiac surgery CABG. PCS=Physical Component Summary. RR=risk ratio. SAQ=Seattle Angina Questionnaire. SF-36=36-item Short Form Health Survey. VAS=visual analogue scale. *Missing data were imputed for the quality-of-life outcomes as per the statistical analysis plan. Actual patient data available for each outcome for the MICS CABG group and the sternotomy CABG group: n=76 in the MICS CABG group and n=77 in the sternotomy CABG group for PCS; n=79 and n=77 for role-physical; n=78 and n=79 for physical functioning; n=79 and n=79 for bodily pain; n=77 and n=79 for general health; n=79 and n=79 for vitality; n=79 and n=79 for social functioning; n=79 and n=78 for role-emotional; n=79 and n=79 for mental health; n=79 and n=80 for EQ-5D VAS; n=79 and n=80 for SAQ; n=76 and n=77 for MCS; n=78 and n=74 for postoperative pain VAS (resting); and n=77 and n=73 for postoperative pain VAS (post-physiotherapy). †Sensitivity analysis comparing MICS CABG and sternotomy CABG using the Wilcoxon rank-sum test: p value 0.048. ‡Covariates: age, sex, use of cardiopulmonary bypass, number of grafts, and BMI. §Wilcoxon rank-sum test.

Table 3: Primary and secondary outcomes at 1 month after surgery

Clinical events to 12 months postoperatively were infrequent in both groups, consistent with observational reports of low adverse-event rates of MICS CABG in patients with multivessel coronary artery disease.^{12,13} The earlier dual-centre STET trial²⁷ used larger anterolateral thoracotomy incisions, and thoracotomy-based surgical revascularisation in this trial resulted in fewer grafts

compared with a sternotomy. Those concerns were not observed in the MIST trial, where the number of grafts, the completeness of revascularisation, and the number of arterial grafts did not differ between the two groups despite a smaller incision (appendix p 22)—suggesting that MICS CABG performed by experienced teams can be done without an apparent reduction in procedural quality or

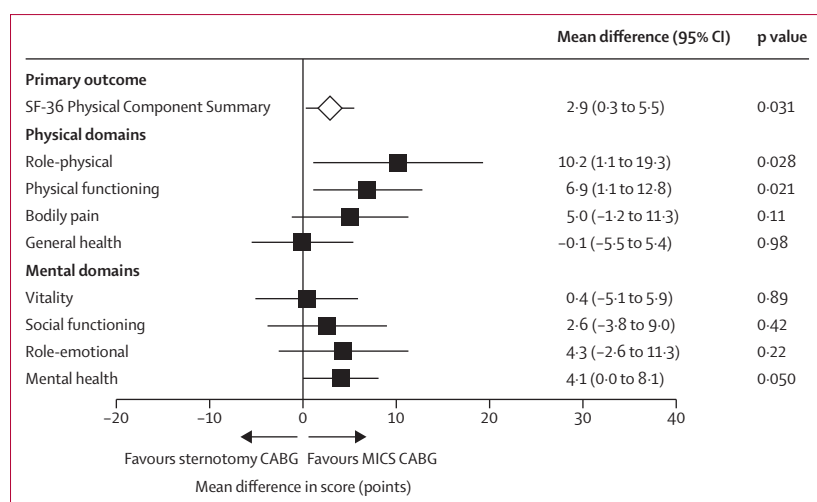


Figure 2: Primary endpoint and SF-36 domain scores at 1 month after surgery

SF-36 scores range from 0 to 100; higher scores indicate better health-related quality of life. Mean differences are MICS CABG minus sternotomy CABG. CABG=coronary artery bypass grafting. MICS CABG=minimally invasive cardiac surgery coronary artery bypass grafting. SF-36=36-item short form health survey.

	MICS CABG group (n=86)	Sternotomy CABG group (n=84)
From surgery to 1 month after surgery		
Death	0	0
MACCE (death, myocardial infarction, stroke, or target vessel revascularisation)	1/86*	0
Re-exploration for bleeding	1/86	3/84
Nosocomial infection with mesenteric ischaemia and cholecystitis	1/86	0
Deep chest wound infection	0	3/84
Haematoma evacuation	0	1/84
Acute on chronic renal failure: need for dialysis	0	1/84
From 1 month to 12 months after surgery†		
Death	0	0
MACCE (death, myocardial infarction, stroke, or target vessel revascularisation)	0	0
Hospital readmission	1/86	5/81
Haemorrhagic pericardial effusion requiring drainage	0	1/81
Repeat admissions for chest wound infection	0	1/81
Mediastinitis	0	1/81
Groin infection requiring vacuum and antibiotics	1/86	0
New renal failure requiring dialysis	0	1/81
Pyelonephritis	0	1/81

MACCE=major adverse cardiac or cerebrovascular event. CABG=coronary artery bypass grafting. MICS CABG=minimally invasive cardiac surgery CABG. *Postoperative catheterisation and re-thoracotomy for kinked graft. †Between 1 month and 12 months postoperatively, three patients in the sternotomy CABG group withdrew consent (n=1) or were lost to follow-up (n=2).

Table 4: Serious adverse events up to 12 months after surgery

arterial graft use. No significant between-group difference was identified in SAQ scores up to 12 months after surgery.

The MIST trial enrolled patients treated by experienced MICS CABG surgeons; most patients were recruited at

two centres (appendix p 2), and each operating surgeon led a dedicated minimally invasive CABG programme. Therefore, the results might not be reproducible in centres or by surgeons who do not have considerable experience in MICS CABG. Nevertheless, the MIST trial provides randomised evidence that MICS CABG can improve early patient-reported physical recovery compared with sternotomy CABG in selected patients treated by experienced teams; subsequent dedicated advanced education, peer-to-peer mentoring, and implementation research could inform whether and how MICS CABG can be adopted more widely—as is the case for other advanced surgical procedures. MICS CABG is part of a broader spectrum of non-sternotomy approaches to CABG, including robotic-assisted coronary surgery (eg, robotic minimally invasive direct coronary artery bypass and totally endoscopic coronary artery bypass) and hybrid coronary revascularisation, intended to decrease surgical invasiveness and improve postoperative patient quality of life while maintaining procedural quality and safety.²⁸

We previously showed that the learning curve for MICS CABG can be addressed by a stepwise approach, including cardiopulmonary bypass support (used in four patients in the MIST trial) when surgical exposure or haemodynamic stability is challenging.^{12,21,29} Cardiopulmonary bypass support allows surgeons to gradually acquire the technical skills required for the operation, with no previously observed differences in clinical events between the initial 25 and subsequent MICS CABG operations. Learning-curve analyses of operative times also suggest that, before initiating off-pump, multivessel MICS CABG, a surgeon and their team should have completed approximately 60 single-vessel MICS CABG and 15 cardiopulmonary bypass-assisted multivessel MICS CABG. Notably, 16% of patients in the MIST trial had preoperative left ventricular dysfunction, with no crossover to sternotomy up to 12 months after surgery and major adverse cardiac events infrequently observed. We consider that left ventricular dilatation, rather than dysfunction alone, often poses the greater technical challenge during MICS CABG, possibly requiring cardiopulmonary bypass support to facilitate grafting of the distal coronary targets. Until a high level of technical and team experience has been achieved, surgeons should be cautious about undertaking MICS CABG in patients with left ventricular dysfunction.

The MIST trial has limitations. During the trial enrolment period, MIST trial surgeons received many direct referrals for MICS CABG, resulting in patients anticipating the MICS CABG operation, with only a small proportion willing and suitable to undergo random assignment. Enrolment and randomisation took longer than expected, partly because of COVID-19 disruptions and because many patients preferred the less invasive approach when offered the two options. Women were particularly under-represented in the trial, which might

reflect perceived technical challenges related to anatomical factors. However, with growing experience in MICS CABG, these barriers can be addressed, expanding the applicability of this approach to a broader range of patients, including women. The results of the MIST trial might inform future studies evaluating less invasive surgical revascularisation, including hybrid coronary revascularisation, in women, for whom the effect of surgical invasiveness on recovery and outcomes remains inadequately studied. Although none were significant, there were some baseline imbalances between groups. Patients could not be masked to the assigned intervention, which might have contributed to bias in patient-reported outcomes. Last, baseline SF-36 scores were not reliably collected before randomisation and were therefore not available for adjustment in the primary analysis, thereby limiting the strength of the trial's conclusions. The heterogeneous clinical status of patients awaiting CABG in the MIST trial, most of whom had received a new diagnosis of severe coronary artery disease, had unstable angina, or presented with an urgent indication for surgery, limited the ability to obtain reliable patient-reported outcomes before randomisation, with some of the baseline conditions (eg, unstable angina) addressed by the trial intervention.

Overall, the effect size of the primary outcome favouring MICS CABG over sternotomy CABG in the MIST trial was modest but consistent with earlier patient-reported physical recovery and differences in prespecified secondary recovery outcomes, including physical domains of the SF-36. Future studies could evaluate patient-reported physical recovery at earlier postoperative timepoints and generate additional clinically meaningful data to optimise recovery (including fast-track discharge pathways and earlier enrolment into cardiac rehabilitation programmes) and resource utilisation after MICS CABG and sternotomy CABG.^{30,31} Follow-up of patients enrolled in the MIST trial up to 5 years is ongoing and will provide insights into the long-term outcomes.

In conclusion, in this multicentre, international, open-label randomised controlled trial comparing MICS CABG with sternotomy CABG for patients with multivessel coronary artery disease, MICS CABG performed by experienced surgeons improved patient-reported physical recovery at 1 month after surgery. Patients in the MICS CABG group received a similar number of grafts, had similar completeness of revascularisation, and had similar arterial graft configuration, with no apparent reduction in procedural quality and no apparent safety penalty through 12 months. These findings support consideration of MICS CABG as a minimally invasive approach to multivessel surgical coronary revascularisation in selected patients treated by experienced teams.

Contributors

MR was the principal investigator. MR, MHG, and GAW designed the trial. MR, SN, AV, PR, KK, PMD, ML, MP, and GAW were involved in

trial conduct and data collection. MR, AS, AB, and YZ were responsible for trial administration. GAW conducted the statistical analysis. MR, MHG, MP, and GAW interpreted the data. MR, MHG, MP, and GAW prepared the manuscript and incorporated further coauthors' inputs into the final manuscript. MR, MHG, YZ, and GAW attest to the accuracy and completeness of the data and the fidelity of the trial to the protocol. All authors provided critical revision of the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication. MR, MHG, YZ, and GAW directly accessed and verified the underlying data reported in the manuscript.

Declaration of interests

MR reports consulting fees from Medtronic. SN reports speaker fees from Medtronic, Edwards, and Intuitive; and advisory board participation for Intuitive. KK reports consulting fees from Johnson & Johnson; and speaker fees from Medtronic and Johnson & Johnson. PMD reports grants or contracts from Atricure; speaker fees from Medtronic, Geister, and Edwards; and is a clinical proctor at Medtronic and Edwards. MAB reports consulting and speaker fees from Edwards Lifesciences, Medtronic, Abbott, and Artivion. RPW reports grants or contracts, consulting fees, or speaker fees from AtriCure, Abbott, Cytosorbents, and Boston Scientific; and participation on a data safety monitoring board or advisory board for Procyon and Artivion. All other authors declare no competing interests.

Data sharing

The study protocol, statistical analysis plan, analytic code, and consent form will be made available for academic purposes upon reasonable request, subject to a data sharing agreement, ethics approval, and protocol approval. Individual participant data underlying the reported results will also be available after de-identification, under the same criteria. Data will be accessible beginning 3 months and ending 5 years after Article publication. Proposals should be directed to MR (mrue@ottawaheart.ca); approved requestors will be required to sign a data access agreement.

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